COMP396 Weekly Meeting Minutes

(Week 4)

Date & Time: Saturday, February 21st , 2026, 13:00–16:00

Location: Tencent Meeting (Online)

Note Taker: Xue Ella (XE)

Attendees: Zhang Olivia (ZO), Zhao Saria (ZS), Xue Ella (XE), Wong Haw (WH)

Note: *Due to ZS's personal health reasons, we were unable to finalize the strategy development plan by the original internal deadline of February 21st. We expect to complete the planning by next Wednesday, after which ZS (lead programmer) will begin the core coding phase for the strategy.*

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1. Agenda

- Review the completion status of the asset-to-strategy mapping.
- Validate data using statistical methodologies and technical tools to evaluate and decide on the final portfolio integration logic, while documenting the rejection of alternative frameworks (MV, CVaR, RP, ML).

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2. Discussion & Decisions

2.1. Final Asset-to-Strategy Mapping

Based on our statistical validation, we have finalized the following mapping of the ten-asset series (Series 01–10) to our chosen strategies:

- Trend/Momentum Strategy**:** Series 01, 03, 04, 06, 09.
- Mean-Reversion Strategy: Series 02, 07, 08, 10.

- Defensive/Short-Selling Strategy: Series 05.

2.2. Quantitative Decision Process and Data Validation

Following WH's proposal to consider portfolio strategies such as Mean-Variance (MV), CVaR, Risk Parity (RP), and Random Forest, ZS performed a series of simulations to evaluate their feasibility. She utilized the following analytical framework to confirm our final structural matching logic:

I. Analytical Framework and Indicators:

- **Data Processing:** We utilized **Pandas** and **NumPy** for time-series computations.
- **Annualised Return:** Calculated as $Annualised\ Return = (\frac{P_r}{P_0})^{\frac{300}{N}}$-1
- **Annualised Volatility:** Calculated as $Ann.Vol = \sigma_{daily} \times \sqrt{300}$
- **Log-Price Trend Regression (OLS):** We used *statsmodels.api.OLS* to estimate

  $$\ln(Pt) = \alpha + \beta t + \varepsilon_t.$$

- **Structural Consistency:** Evaluated via the SMA200 Ratio: $\frac{Number\ of\ days\ P_t > SMA200_t}{Total\ valid\ days}$

II. Statistical Evidence Table (Simulated by ZS):

Series	Total Return	Ann. Vol	Trend t-stat (|t| > 2)	R^2	ADF p-value	Above SMA200	Mapping Result
1	High	High	Sig. Positive	High	High	High	Trend/Momentum
2	Low	High	Insignificant	Low	Relatively Low	Medium	Mean-Reversion
3	Medium	Low	Sig. Positive	Medium	High	High	Trend/Momentum
4	Extreme	Extreme	Sig. Positive	High	High	High	Trend/Momentum
5	Negative	Medium	Sig. Positive	High	High	Low	Defensive/Short
6	High	Medium	Pos. but Weak	Medium	High	Medium-High	Trend/Momentum

7	Low	Low	Insignificant	Low	Low	Medium	Mean-Reversion
8	Marginal	Low	Insignificant	Low	Low	Medium	Mean-Reversion
9	Medium	Medium	Sig. Positive	Medium	High	High	Trend/Momentum
10	Medium	High	Weak Positive	Low	Medium	Medium	Mean-Reversion

III. Strategic Rationale for Allocation:

- **Trend Assets:** Series 1, 3, 4, 6, and 9 exhibit positive OLS slopes with statistically significant t-stats. High ADF p-values indicate non-stationary behavior (random walk with drift), making them ideal for trend continuation logic.

- **Mean-Reversion Assets:** Series 2, 7, 8, and 10 show low R^2 and insignificant slopes. Low ADF p-values suggest price action is closer to a stationary process, supporting a mean-reversion hypothesis.

- **Defensive Assets:** Series 5 exhibits a statistically significant negative trend slope in the regression model. The price remains predominantly below the 200-day moving average, confirming a persistent downward structural regime. Combined with negative cumulative return and downside risk characteristics, this series is more appropriately aligned with defensive or short-based strategies.

2.3. Rejected Methodologies and Mitigation

Based on the simulation results, we decided to reject the following methodologies:

- **MVO & CVaR:** Mean-Variance and CVaR frameworks operate at the portfolio allocation level and rely on stable estimates of expected returns and risk matrices. Given the limited number of assets and sample size, covariance estimation may be unstable and highly sensitive to input noise. Moreover, high-volatility but structurally trending assets could be systematically underweighted under variance- or tail-risk-based optimization, potentially suppressing structural alpha. Since the research objective focuses on structural strategy matching rather than risk minimization, these methods were not adopted.

- **Risk Parity (RP):** RP aims to equalize risk contributions across assets, assuming relatively homogeneous structural characteristics. However, the dataset exhibits clear heterogeneity in trend persistence and stationarity. Applying a uniform risk-balancing framework may dilute the contribution of structurally trending assets and weaken regime-specific strategy effects. Therefore, Risk Parity was considered inconsistent with the study's structural classification approach.

- **Machine Learning (Random Forest):** Random Forest and other machine learning models require

larger sample sizes and extensive feature engineering to achieve robust performance. In the current setting, introducing high-dimensional models would increase complexity and overfitting risk while reducing interpretability. Given that statistical diagnostics already provide clear structural differentiation, a parsimonious and interpretable framework was deemed more appropriate.

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3. Task Allocation

Following our decision, we have assigned the following responsibilities for the evaluation phase:

ID	Task Description	Owner	Due Date
C1	Formulate the strategy development plan	All members	Feb 25
C2	Start writing the code according to the plan and report on the progress weekly	ZS	—
C3	Formulate the plan for optimization and start to write the first part of the final report (why choose these strategies)	WH & ZO & XE	—

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4. Current Challenges and Issues

- **Timeline Compression:** Because our development plan was delayed, we must accelerate our logic validation to ensure ZS can begin programming the T-Strategy

class on schedule next week.

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5. Next Meeting Arrangement

- **Date & Time:** Thursday, February 26th, 2026, 17:00–19:00
- **Location:** Harold Cohen Library
- **Key Agenda:**
  - Finalize the programming plan.
  - ZS gives a code progress report and other team members offer their opinions.